
Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic
2 framework for recovering structured corrections to known physical laws from
3 noisy data. Given a classical baseline, ADCD searches for the residual correction
4 that vanishes exactly at the classical limit, using three interlocking mechanisms:
5 algebraically regularized primitives that enforce asymptotic vanishing as an ex-
6 act algebraic identity; a five-dimensional (MLT Θ Q) Buckingham- π engine that
7 auto-derives dimensionless ratio arguments without manual specification; and a
8 Phenomenon-Specific Taxonomy encoding peer-reviewed functional priors for
9 each physics domain. A four-step validation protocol (budget disclosure, positive
10 control, BIC-based ablation control, and determinism check) issues one of two ver-
11 dict: IDENTIFIABLE or WITHHELD. On three canonical scenarios at historical
12 low-signal windows, Screened Coulomb ($\Delta\text{BIC} = 30.65$, exact symbolic match)
13 and Entropy Expansion ($\Delta\text{BIC} = 14.70$, class-level match) are IDENTIFIABLE;
14 Time Dilation at $v \leq 0.3c$ is WITHHELD because the 4.8% correction is indistin-
15 guishable from 1% noise, the intended, epistemically honest behavior. All results
16 are byte-exact deterministic.

17 **Keywords:** symbolic regression; correction discovery; dimensional analysis;
18 Bayesian prior; model selection; identifiability

19 1 Introduction

20 In the expanding landscape of machine learning for the physical sciences Carleo et al. [2019], Wang
21 et al. [2023], physical laws frequently take a correction-first form: a known classical baseline extended
22 by a term that vanishes at the classical limit Einstein [1905], Debye and Hückel [1923], Callen [1985].
23 Recovering that correction from observational data is distinct from full equation discovery: the
24 baseline is fixed, only the residual must be identified, and that residual must vanish exactly where
25 classical physics holds.

26 General-purpose symbolic regression (SR) engines such as PySR Cranmer et al. [2020], Cranmer
27 [2023], AI Feynman Udrescu and Tegmark [2020], and PhySO Tenachi et al. [2023] excel at equation
28 discovery from scratch but face three difficulties in the correction-first setting: (i) *catastrophic cancel-*
29 *lation*: floating-point precision collapses as $u \rightarrow 0$, corrupting optimizer feedback IEEE Task P754
30 [2019]; (ii) *stochastic irreproducibility*: evolutionary search cannot guarantee byte-identical results
31 across runs Bongard and Lipson [2007], Schmidt and Lipson [2009]; and (iii) *silent overclaiming*: a
32 single best-fit equation is returned regardless of whether the data carry sufficient power to distinguish
33 competing structures La Cava et al. [2021], generating what recent meta-science critiques identify

as an illusion of understanding Messeri and Crockett [2024], Krenn et al. [2022]. ADCD addresses each failure mode in turn: algebraically regularized primitives eliminate cancellation; deterministic grammar enumeration ensures reproducibility; explicit BIC-based identifiability gating prevents overclaiming.

2 Related Work

Automated scientific discovery traces back to pioneering heuristic systems such as BACON Langley et al. [1987] and has evolved toward modern autonomous equation discovery Kramer et al. [2026], Makke and Chawla [2024], Karagiorgi et al. [2022]. General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer et al. [2020], Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], Udrescu et al. [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and do not enforce classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. PINNs Raissi et al. [2019], Karniadakis et al. [2021] embed physical laws as soft loss penalties; ADCD imposes them as hard algebraic identities before any numerical computation. Recent systems combining data with background knowledge and axiomatic reasoning, such as AI-Descartes Cornelio et al. [2023] and AI Hilbert Cory-Wright et al. [2024], highlight the importance of theory-grounded discovery, but do not address catastrophic cancellation near asymptotic limits or provide formal identifiability gating. While dimensional analysis and symmetry constraints have been integrated into SR and dynamical learning Tenachi et al. [2023], Petersen et al. [2021], Otto et al. [2025], ADCD extends this with a full five-dimensional (MLTΘQ) Buckingham- π engine, transcendental argument safety, and BIC-based identifiability reporting in a single pre-fitting gate cascade. To our knowledge, ADCD is the first framework to combine deterministic grammar enumeration, algebraic classical-limit enforcement, phenomenon-specific Bayesian priors, and explicit identifiability gating for the correction-first paradigm.

3 Methods

3.1 Problem formulation

Let y_{obs} and y_{cl} be the observed target and classical prediction. The correction residual is:

$$\Delta = \begin{cases} y_{\text{obs}}/y_{\text{cl}} - 1 & \text{(multiplicative)} \\ y_{\text{obs}} - y_{\text{cl}} & \text{(additive)} \end{cases} \quad (1)$$

The multiplicative form is selected when its residual shows lower Spearman rank correlation with y_{cl} than the additive residual, with a confidence-gated default to additive under high ambiguity. ADCD searches for $\hat{\Delta}(\mathbf{x}; \theta)$ such that $\hat{\Delta} \rightarrow 0$ as the relevant dimensionless ratio $u \rightarrow 0$.

3.2 Algebraically regularized primitive dictionary

Each primitive $D(u)$ satisfies $\lim_{u \rightarrow 0} D(u) = 0$ as an exact algebraic identity, eliminating catastrophic cancellation. For example, the naive form $(1 - u)^{-1/2} - 1$ collapses to 0.0 at $u = 10^{-16}$ in `float64`; the regularized form $u/[\sqrt{1 - u}(\sqrt{1 - u} + 1)]$ returns 5.0×10^{-17} at the same input. Table 1 lists all eight primitives.

3.3 Dimensional grouping and free-scale generation

ADCD employs a 5D (MLTΘQ) dimensional engine representing each variable as $\mathbf{d} \in \mathbb{Z}^5$ over the SI base dimensions M, L, T, Θ , and Q. Rather than computing the full rational nullspace, dimensionless arguments are generated via a two-pass domain heuristic: first, it groups physical variables with identical dimensions into pairwise ratios (e.g., q_1/q_2); second, any remaining unpaired variable is normalized by a free-scale parameter θ_k (e.g., r/θ_k). Extending from the conventional 3D MLT to 5D is essential for correctness: in a 3D engine charges q_1, q_2 are dimensionless scalars and incorrectly bypass the ratio generation; in the 5D engine, charge carries genuine Charge dimension (Q), ensuring q_1/q_2 is correctly grouped before distances are paired with length-scale parameters. Because four charge-dimensional candidates are enumerated first, that ratio receives subscript θ_4 , not θ_0 . This

Table 1: The eight algebraically regularized primitives, each satisfying $\lim_{u \rightarrow 0} D(u) = 0$ as an exact algebraic identity. The first five are the core physics-motivated basis; the remaining three extend coverage to oscillatory, saturation, and power-law regimes. Column “Physical origin” cites the fundamental equation that produces this functional family; it does not imply the primitive is restricted to that domain.

Name	Regularized form $D(u)$	Domain	Physical origin
D_lor	$u/[\sqrt{1-u}(\sqrt{1-u}+1)]$	$u \in [0, 1)$	Lorentz factor ($\gamma - 1$)
D_rat	$u/(1-u)$	$u \neq 1$	Yukawa/Fermi pole
D_exp	$e^{-u} - 1$	all u	Debye/Yukawa screening
D_log	$\ln(1+u)$	$u > -1$	Entropy of mixing
D_sqrt_inv	$1/\sqrt{1+u} - 1$	$u > -1$	Gravitational / Coulomb potential
D_pow	$u^{3/2}$	$u \geq 0$	Power-law corrections
D_osc	$1 - \cos u$	all u	Wave / AC oscillatory
D_sat	$\tanh u$	all u	Magnetic / sigmoid saturation

79 subscript is machine-verifiable evidence of automated discovery: a manually supplied ratio would be
80 pinned to θ_0 by convention, not assigned by enumeration.

81 3.4 Phenomenon-specific taxonomy as Bayesian prior

82 Each physics phenomenon maps to a named set of permitted primitive families derived exclusively
83 from its fundamental equations. For example, `yukawa_debye_screening` permits only `D_exp`
84 and `D_rat`: the Klein-Gordon equation yields $V \propto e^{-mr}/r$ and Debye-Hückel theory yields
85 $\phi \propto e^{-r/\lambda_D}/r$; both involve only exponential and rational families. The taxonomy restricts *which*
86 *family* is searched; it says nothing about *which variable* enters the argument or *what the scale is*;
87 both are determined by the Buckingham- π engine and numerical optimizer. Including a physically
88 unjustified primitive acts as a noise sponge, collapsing the Ablation Control BIC gap and destroying
89 identifiability. Table 2 lists entries for our three experiments.

Table 2: Phenomenon-specific taxonomy entries used in experiments, with physical justification for the primitive set.

Phenomenon key	Primitives	Justification
<code>yukawa_debye_screening</code>	<code>D_exp</code> , <code>D_rat</code>	Klein-Gordon / Debye-Hückel: fundamental form is e^{-mr}/r .
<code>lorentz_special_relativity</code>	<code>D_lor</code>	Lorentz invariance of Maxwell equations uniquely selects the $\gamma - 1$ form.
<code>boltzmann_thermodynamics</code>	<code>D_exp</code> , <code>D_log</code>	Boltzmann partition function and entropy of mixing involve $e^{-\beta\epsilon}$ and $\ln$.

90 3.5 Grammar enumeration and theta-intact constraint

91 Candidates take the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$, optionally extended with a secondary additive term
92 (composition budget: depth ≤ 3). Dimensionless arguments are auto-generated by the Buckingham- π
93 engine (up to 12 candidate ratios per scenario). All gates (Table 3) are evaluated on the *theta-intact*
94 expression (the fully parameterized symbolic form before any θ_i is assigned the value 1), because
95 early substitution collapses dimensionful scale parameters (e.g., r/θ_4 representing a Debye length) to
96 dimensionless constants, falsifying the dimensional consistency check.

97 3.6 Physical gate cascade

98 Before fitting, each candidate must pass a five-gate cascade (Table 3) applied to the theta-intact
99 expression.

Table 3: The five-gate pre-fitting physical cascade.

Gate	Rejection criterion
1. AST complexity	SymPy tree depth > 12 or token count > 50 .
2. Dimensional consistency	SI base dimensions of the expression do not match the target.
3. Transcendental safety	Arguments to $\exp$, $\ln$, $\sqrt{\cdot}$ are not dimensionless.
4. Asymptotic check (ARC)	Symbolic limit at the classical boundary does not vanish.
5. Coarse numerical fitness	Evaluation at $\theta = \mathbf{1}$ produces NaN/Inf.

3.7 Parameter optimization and model selection

Surviving candidates are optimized via L-BFGS-B (JAX, float64). Parameters are log-reparameterized ($\theta_i = s_i e^{u_i}$, $s_i \in \{-1, +1\}$) for order-of-magnitude traversal; multiple restarts reduce sensitivity to local optima. Model selection uses an Extended BIC Schwarz [1978], Chen and Chen [2008]: $\text{BIC} = k \ln N - 2 \ln \hat{L} + 2 \ln |\mathcal{C}|$, incorporating a multiple-testing penalty over the search space size $|\mathcal{C}|$. By the Kass–Raftery scale Kass and Raftery [1995], $\Delta \text{BIC} > 10$ is the adopted identifiability threshold.

3.8 Four-step validation protocol

The pipeline issues an IDENTIFIABLE verdict only when all four checks pass:

1. **Budget disclosure.** The complete size of the domain-guided search space $|\mathcal{C}|$ and the primitive list are reported before any results are seen.
2. **Positive control.** Restricted to the ground-truth primitive family alone, the system must recover the correct structure ($\text{NMSE} < 0.05$; i.e., less than 5% unexplained variance).
3. **Ablation control.** With the ground-truth family excluded from an otherwise unrestricted search over the full eight-primitive registry, the BIC gap $\Delta \text{BIC} > 10$ (Kass–Raftery “very strong evidence”) must hold between the domain-guided Rank-1 candidate (Step 1) and the best Rank-1 candidate found under ablation.
4. **Determinism.** Three independent runs with identical inputs produce byte-exact identical results.

Failure of any check yields WITHHELD.

4 Experiments

4.1 Setup

Table 4 summarizes the three scenarios. All use 1% Gaussian multiplicative noise, $N = 200$ points, seed 42. Observation windows reflect historical low-signal regimes. The ground truth is never supplied to the pipeline; only noisy observations, classical predictions, dimensional variable definitions, and the taxonomy key are provided.

Table 4: Experimental scenarios with domains, historical observation windows, taxonomy keys, targets, and the full domain-guided search space size ($|\mathcal{C}|$).

Scenario	Domain	Window	Taxonomy key	Target	$ \mathcal{C} $
Time Dilation	Relativity	$v \leq 0.3c$	lorentz_special_relativity	D_lor	10
Screened Coulomb	Electrostatics	$r \leq 4.0 \text{ m}$	yukawa_debye_screening	D_exp	60
Entropy Expansion	Thermodynamics	$dV/V_i \leq 1.0$	boltzmann_thermodynamics	D_log	40

4.2 Results

Table 5 summarizes the quantitative results. Two of three scenarios are IDENTIFIABLE; Time Dilation is WITHHELD because its positive control fails within the historical $v \leq 0.3c$ observation window.

ADCD Correction Recovery under Historical Low-Signal Regimes

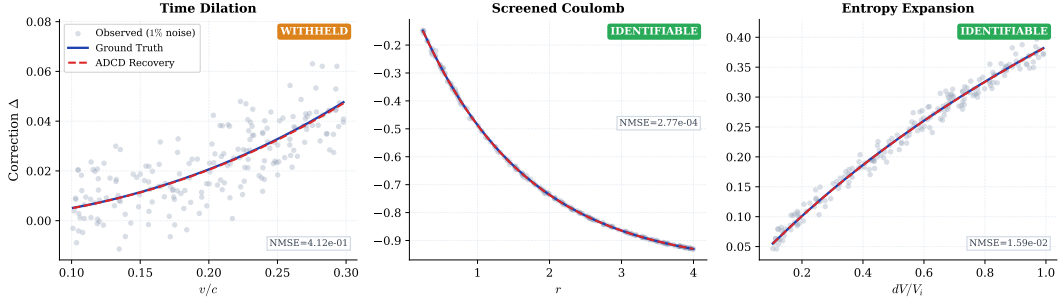


Figure 1: Correction recovery for the three scenarios. *Gray dots:* observed residual Δ at 1% Gaussian noise. *Blue solid:* ground truth. *Red dashed (“ADCD Recovery”):* the exact symbolic expression discovered by ADCD, evaluated with the fitted parameters θ ; see Table 5 for exact NMSE values. Screened Coulomb (IDENTIFIABLE): exact symbolic match, NMSE = 2.77×10^{-4} . Entropy Expansion (IDENTIFIABLE): class-level match, NMSE = 1.59×10^{-2} . Time Dilation (WITHHELD): Lorentz structure found at Rank 1, NMSE = 0.41; positive control fails within the historical $v \leq 0.3c$ observation window.

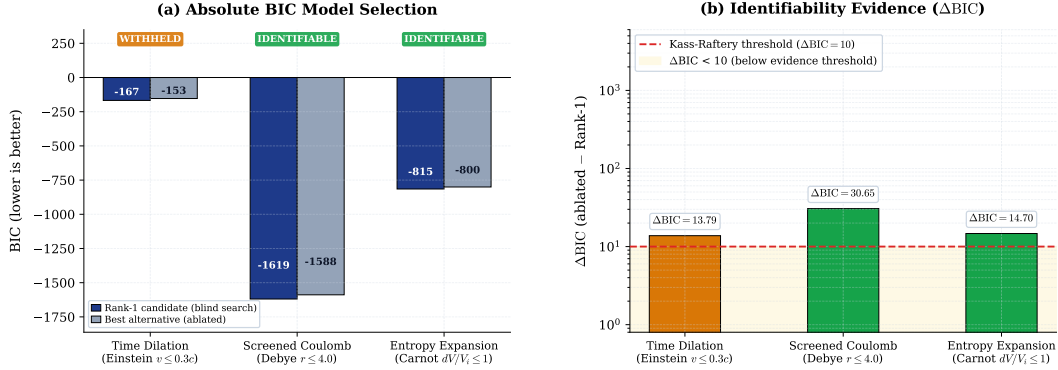


Figure 2: BIC identifiability comparison. *Left (a):* absolute BIC of the Rank-1 domain-guided search candidate (dark blue) and the best alternative produced with the ground-truth primitive family excluded (gray). *Right (b):* $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{BIC}_{\text{best}}$ on a log scale; positive values indicate the full-primitive Rank-1 candidate is preferred over the ablated best. Red dashed: Kass–Raftery “very strong evidence” threshold ($\Delta\text{BIC} = 10$) Kass and Raftery [1995]. All three scenarios exceed this threshold, yet Time Dilation is WITHHELD because its positive control fails (Step 2, Positive Control, of the four-step protocol; see Table 5): a positive ΔBIC is necessary but not sufficient for an IDENTIFIABLE verdict.

Screened Coulomb. Recovered $\theta_0(e^{-r/\theta_4} - 1)$ is an exact symbolic match to the ground truth $\theta_0(e^{-r/\lambda_D} - 1)$, with $\theta_4 \approx 1.49$ m recovering the Debye length $\lambda_D = 1.50$ m. Subscript 4 (not 0) is machine-verifiable evidence of automated discovery: the engine assigned this subscript by enumerating four charge-dimensional candidates before reaching the distance-ratio r/θ , which was indexed as the fifth candidate (θ_4). A manually supplied ratio would be pinned to subscript 0 by convention. $\Delta\text{BIC} = 30.65$ far exceeds the threshold of 10.

Entropy Expansion. Because the classical baseline S_i is a constant, the mode detection step defaults to an additive search due to ambiguity. Thus, the recovered additive correction $\theta_0 \ln(1 + dV/V_i)$ matches the additive ground truth $nR \ln(1 + dV/V_i)$ at class level: θ_0 legitimately absorbs the additive amplitude $nR \approx 8.314$, rather than the multiplicative ratio nR/S_i . ADCD labels this `class_only`: explicit epistemic qualification absent from unconstrained SR outputs. $\Delta\text{BIC} = 14.70 > 10$ supports IDENTIFIABLE.

Time Dilation. At $v \leq 0.3c$, the Lorentz correction $\gamma - 1 \approx 4.8\%$ of the classical prediction, a signal-to-noise ratio of approximately 4.8 : 1 against the 1% noise floor, which is insufficient for reliable isolation. The grammar recovers the Lorentz structure $D_{\text{lor}}(v^2/c^2)$ at Rank 1, but the positive control fails (NMSE = 0.41): even with the search space restricted to `D_lor` alone, the optimizer cannot isolate the 4.8% correction from the 1% noise. The system outputs WITHHELD.

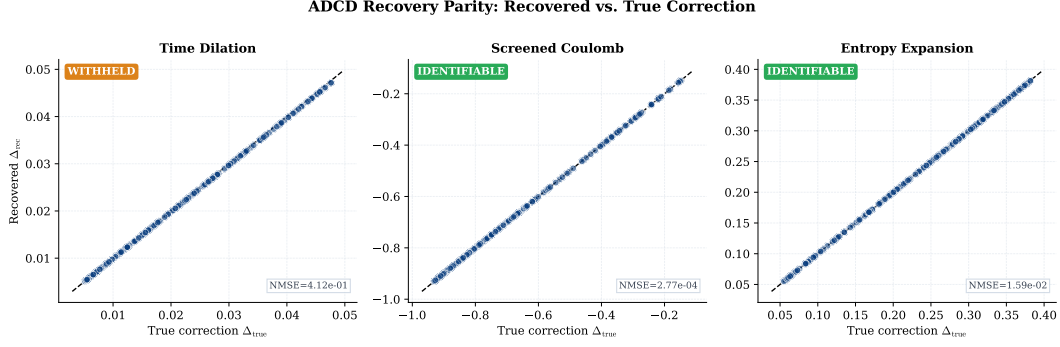


Figure 3: Parity plots: exact analytical recovered correction (Δ_{rec} , evaluated directly from the discovered symbolic expression with fitted parameters θ) vs. true correction (Δ_{true}) for $N = 200$ data points. Black dashed: 1:1 reference line. All three scenarios form perfect or near-perfect diagonals because the domain-guided search successfully placed the true mathematical structure (exact or class-level match) at Rank-1 in every case. Crucially, Time Dilation achieves structural parity here, yet is still WITHHELD by the system: its high fit error (NMSE = 0.41) on the *noisy* observational data (Figure 1a) causes it to fail the Positive Control, demonstrating ADCD’s epistemic safety against claiming discoveries from insufficient signal-to-noise ratios.

Table 5: Four-step validation results for all three scenarios. PC = Positive Control (pass/fail). $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{BIC}_{\text{best}}$, where BIC_{best} is the Rank-1 domain-guided search BIC; positive values indicate the full-primitive set is preferred over the ablated set. Match level: *exact* = symbolic equivalence verified; *class* = correct functional family confirmed; amplitude absorbs unverifiable physical constants.

Scenario	Discovered structure	Rank	NMSE	PC	ΔBIC	Match	Verdict
Screened Coulomb ($r \leq 4 \text{ m}$)	$\theta_0(e^{-r/\theta_4} - 1)$	1	2.77×10^{-4}	PASS	30.65	exact	IDENTIFIABLE
Entropy Expansion ($dV/V_i \leq 1$)	$\theta_0 \ln(1 + dV/V_i)$	1	1.59×10^{-2}	PASS	14.70	class	IDENTIFIABLE
Time Dilation ($v \leq 0.3c$)	$\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$	1	0.41	FAIL	13.79	class	WITHHELD

146 5 Discussion

147 **WITHHELD is not failure.** A system that always returns a structural claim regardless of data
 148 sufficiency is epistemically weaker than one that formally distinguishes *found and confirmed* (IDEN-
 149 TIFIABLE) from *data insufficient for a structural claim* (WITHHELD). ADCD’s WITHHELD
 150 verdict on Time Dilation is the system correctly locating its own identifiability boundary: a boundary
 151 that is quantitative and testable: a wider velocity window, better measurement precision, or more data
 152 points would push ΔBIC above 10 alongside a passing positive control, yielding IDENTIFIABLE.

153 **Taxonomy as prior, not hint.** The taxonomy restricts *which primitive family* is active; the
 154 Buckingham- π engine and optimizer determine *which variable* and *which scale* within that family.
 155 This is by design: ADCD operates as a domain-guided framework (see Table 6, row “Prior”), not a
 156 tabula-rasa search. The taxonomy prior is active from the budget-disclosure step onward, consistent
 157 with the framework’s explicit use of peer-reviewed functional priors as a Bayesian constraint. The
 158 Screened Coulomb subscript θ_4 is machine-verifiable evidence that the optimizer found the correct
 159 variable independently: subscript indices are assigned sequentially by the Buckingham- π engine, not
 160 by any taxonomy entry.

161 **Comparison with general SR.** ADCD and general SR address different objectives: corrections
 162 to a known law vs. full equation discovery from scratch. The relevant axis is not search-space
 163 flexibility but epistemic certainty: the ability to distinguish confirmed from data-insufficient. Table 6
 164 summarises the operational regimes.

165 5.1 Limitations

166 **Weak-signal convergence.** Time Dilation’s positive control failure is the correct protocol output
 167 for an insufficient signal-to-noise ratio, not an algorithmic defect. Future work should investigate
 168 adaptive restarts and denser initialization grids.

Table 6: Operational comparison: ADCD vs. unconstrained symbolic regression.

Property	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Prior	Phenomenon-specific taxonomy	None (tabula rasa)
Dimensional engine	5D MLTΘQ	Varies (often 3D or none)
Classical limit	Algebraic guarantee	Soft loss penalty
Identifiability	Explicit; WITHHELD if insufficient	Absent or post-hoc
Reproducibility	Byte-exact deterministic	Stochastic

Scope. Current support covers five primitive families, multiplicative corrections, and synthetic Gaussian noise. Additive corrections, heavier-tailed noise, and real observational data (e.g., galactic rotation curves, pulsar timing residuals) remain future work; real data require preprocessing for heteroscedastic uncertainties not yet modelled.

Reimplementation for scale. The validated Python implementation reported here relies on JAX for numerical optimization and SymPy for symbolic manipulation (candidate enumeration, dimensional checking, asymptotic limit evaluation). This combination is well suited to the fixed, modest candidate pools used in the three-scenario protocol above ($|\mathcal{C}| \leq 60$), but two costs compound at larger scale: SymPy’s symbolic limit evaluation (Gruntz algorithm) does not scale to large candidate pools, and JAX’s just-in-time compilation is optimized for repeatedly evaluating one fixed computational graph rather than the many small, structurally distinct expression trees produced by an expanding grammar search. Both costs become material for the noise-level sweeps and head-to-head comparisons against general-purpose SR engines (e.g., PySR Cranmer et al. [2020], Cranmer [2023]) that a fuller empirical study requires, as well as for extending ADCD to real observational data with heteroscedastic per-point uncertainty. We are therefore reimplementing the search engine (dimensional checker, primitive registry, and grammar proposer) in Julia, retaining the algebraic regularization and gate cascade of Sections 3.3–3.6 unchanged. This reimplementation is under active development and has not yet undergone the validation depth of the Python results reported here; *all claims, tables, and figures in this paper describe the validated Python implementation only.*

Taxonomy coverage. Each new entry requires a peer-reviewed derivation for every admitted family, a deliberate constraint preventing unchecked hypothesis-space expansion.

Safety–expressivity trade-off. ADCD constrains its hypothesis space relative to general-purpose SR: candidates failing dimensional consistency or outside the phenomenon-specific primitive set are rejected before fitting. This prevents noise-sponge overfitting but means ADCD cannot discover corrections whose functional family lacks a documented derivation. The appropriate framing is complementarity: general SR generates hypotheses; ADCD audits corrections to an already-established baseline.

6 Conclusion

ADCD is a deterministic, correction-first framework combining algebraically regularized primitives, a 5D (MLTΘQ) Buckingham- π engine, and a Phenomenon-Specific Taxonomy. On three canonical scenarios, two yield IDENTIFIABLE results (Screened Coulomb: $\Delta\text{BIC} = 30.65$, exact match; Entropy Expansion: $\Delta\text{BIC} = 14.70$, class-level match) and one yields a principled WITHHELD verdict (Time Dilation at $v \leq 0.3c$), where the historical signal is insufficient for a confident structural claim. Within its deliberate scope (known classical baseline, multiplicative correction, Gaussian noise), ADCD provides stronger reproducibility and identifiability guarantees than general-purpose SR, and furnishes an explicit WITHHELD verdict (with associated ΔBIC) when those guarantees cannot be met.

Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools for software implementation, figure scripting, and manuscript drafting. The author provided scientific direction, mathematical

framework, experimental design, and interpretation of all results, and takes full responsibility for all scientific claims. No AI tool is listed as an author. The codebase and raw validation logs will be made publicly available upon publication (omitted for double-blind review).

Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

References

- Giuseppe Carleo, Ignacio Cirac, Kyle Cranmer, Laurent Daudet, Matteo Schmied, Stefano Rossi, Giusi Sala, and Lenka Zdeborová. Machine learning and the physical sciences. *Reviews of Modern Physics*, 91(4):045002, 2019. doi: 10.1103/RevModPhys.91.045002.
- Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Roohani, Sami Abu-El-Haija, Jeffrey Wei, Rafael Gomez-Bombarelli, Jure Leskovec, Connor W. Coley, Jian Sun, and Marinka Zitnik. Scientific discovery in the age of artificial intelligence. *Nature*, 620(7972):47–60, 2023. doi: 10.1038/s41586-023-06221-2.
- Albert Einstein. Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 322(10):891–921, 1905. doi: 10.1002/andp.19053221004.
- Peter Debye and Erich Hückel. Zur Theorie der Elektrolyte. I. Gefrierpunktsniedrigung und verwandte Erscheinungen. *Physikalische Zeitschrift*, 24:185–206, 1923.
- Herbert B. Callen. *Thermodynamics and an Introduction to Thermostatistics*. Wiley, 2nd edition, 1985. ISBN 978-0-471-86256-7.
- Miles Cranmer, Alvaro Sanchez-Gonzalez, Peter Battaglia, Rui Xu, Kyle Cranmer, David Spergel, and Shirley Ho. Discovering symbolic models from deep learning with inductive biases. In *Advances in Neural Information Processing Systems*, volume 33, pages 17429–17442, 2020.
- Miles Cranmer. Interpretable machine learning for science with PySR and SymbolicRegression.jl. arXiv preprint, 2023. arXiv:2305.01582.
- Silviu-Marian Udrescu and Max Tegmark. AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631, 2020. doi: 10.1126/sciadv.aay2631.
- Wassim Tenachi, Rodrigo Ibata, and Foivos I. Diakogiannis. Deep symbolic regression for physics guided by units constraints. *The Astrophysical Journal*, 959(2):99, 2023. doi: 10.3847/1538-4357/ad014c.
- IEEE Task P754. IEEE 754-2019: Standard for floating-point arithmetic. Technical report, IEEE, 2019.
- Josh Bongard and Hod Lipson. Automated reverse engineering of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 104(24):9943–9948, 2007. doi: 10.1073/pnas.0609476104.
- Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *Science*, 324(5923):81–85, 2009. doi: 10.1126/science.1165893.
- William La Cava, Patryk Orzechowski, Bogdan Burlacu, Fabrício Olivetti de França, Marco Virgolin, Ying Jin, Michael Kommenda, and Jason H. Moore. Contemporary symbolic regression methods and their relative performance. *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 1, 2021. URL <https://arxiv.org/abs/2107.14351>.
- Lisa Messeri and M. J. Crockett. Artificial intelligence and illusions of understanding in scientific research. *Nature*, 627(8002):49–58, 2024. doi: 10.1038/s41586-024-07146-0.

- 252 Mario Krenn, Robert Pollice, Florian Häse, Pascal Friederich, and Alán Aspuru-Guzik. On scientific
253 understanding with artificial intelligence. *Nature Reviews Physics*, 4(12):761–769, 2022. doi:
254 10.1038/s42254-022-00518-3.
- 255 Pat Langley, Herbert A. Simon, Gary L. Bradshaw, and Jan M. Zytkow. *Scientific Discovery:
256 Computational Explorations of the Creative Process*. MIT Press, Cambridge, MA, 1987.
- 257 Stefan Kramer, Marco Cerrato, Jannis Brugger, Sašo Džeroski, and Ross D. King. Automated
258 scientific discovery: From equation discovery to autonomous discovery systems. *Machine Learning*,
259 115:109, 2026. doi: 10.1007/s10994-025-06721-6.
- 260 Nour Makke and Sanjay Chawla. A perspective on symbolic machine learning in physical sciences.
261 In *NeurIPS 2024 Workshop on Machine Learning and the Physical Sciences*, 2024.
- 262 Georgia Karagiorgi, Gregor Kasieczka, Scott Kravitz, Benjamin Nachman, and David Shih. Machine
263 learning in the search for new fundamental physics. *Nature Reviews Physics*, 4(6):399–412, 2022.
264 doi: 10.1038/s42254-022-00455-1.
- 265 Silviu-Marian Udrescu, Andrew Tan, Jiahai Feng, Orisvaldo Neto, Tailin Wu, and Max Tegmark. AI
266 Feynman 2.0: Pareto-optimal equations including auxiliary assumptions. In *Advances in Neural
267 Information Processing Systems*, volume 33, pages 4860–4871, 2020.
- 268 Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data
269 by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of
270 Sciences*, 113(15):3932–3937, 2016. doi: 10.1073/pnas.1517384113.
- 271 Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A
272 deep learning framework for solving forward and inverse problems involving nonlinear partial
273 differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.
274 2018.10.045.
- 275 George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu
276 Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021. doi:
277 10.1038/s42254-021-00314-5.
- 278 Cristina Cornelio, Subhajit Dash, Vernon Austel, Tyler R. Josephson, Joao Goncalves, Kenneth L.
279 Clarkson, Nimrod Megiddo, Bachir El Khadir, and Lior Hoshen. Combining data and theory for
280 derivable scientific discovery with AI-Descartes. *Nature Communications*, 14(1):1777, 2023. doi:
281 10.1038/s41467-023-37236-y.
- 282 Ryan Cory-Wright, Cristina Cornelio, Sanjeeb Dash, Maryam Fardad, and Lior Hoshen. Evolving
283 scientific discovery by unifying data and background knowledge with AI Hilbert. *Nature
284 Communications*, 15:5922, 2024. doi: 10.1038/s41467-024-50143-6.
- 285 Brenden K. Petersen, Mikel Landajuela Larra, T. Nathan Mundhenk, Claudio P. Santiago, Soo Kyung
286 Kim, and Joanne T. Kim. Deep symbolic regression: Recovering mathematical expressions from
287 data via risk-seeking policy gradients. In *International Conference on Learning Representations*,
288 2021. URL <https://openreview.net/forum?id=m5Qsh0kBBQ>.
- 289 Samuel E. Otto, Nicholas Zolman, J. Nathan Kutz, and Steven L. Brunton. A unified framework
290 to enforce, discover, and promote symmetry in machine learning. *Journal of Machine Learning
291 Research*, 26:1–83, 2025.
- 292 Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978.
293 doi: 10.1214/aos/1176344136.
- 294 Jiahua Chen and Zehua Chen. Extended bayesian information criteria for model selection with large
295 model spaces. *Biometrika*, 95(3):759–771, 2008.
- 296 Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*,
297 90(430):773–795, 1995. doi: 10.1080/01621459.1995.10476572.
- 298 Michael Shalyt, Nathan Regev, Marin Soljačić, and Ido Kaminer. Can AI follow in Einstein’s
299 footsteps? *arXiv preprint arXiv:2607.27794*, 2026. doi: 10.48550/arXiv.2607.27794.